

Executive Hotel Booking EDA Report

Day 15 — End-to-End Exploratory Data Analysis

Executive Summary

The dataset contains **25,180** booking records and **36** columns. The analysis covers data quality, cleaning, descriptive statistics, univariate and bivariate analysis, group-wise comparisons, correlation analysis, and management recommendations.

Data Quality Assessment

Duplicate rows identified: **180**.

Column	Missing	Missing %
Company_ID	20,729	82.32%
Country	466	1.85%
Agent_ID	304	1.21%
Meal_Type	245	0.97%
Satisfaction_Score	164	0.65%
Children	121	0.48%
Hotel_Location	106	0.42%
ADR	82	0.33%

Key Business Insights

- Overall cancellation rate is 72.2% of bookings.
- Lead time and cancellation have a correlation of 0.34; longer booking lead times are associated with higher cancellation likelihood in this dataset.
- ADR and cancellation have a correlation of -0.02.
- Resort Hotel has the highest average ADR among hotel types, while CITY HOTEL has the highest cancellation rate.
- Online TA is the largest market segment by booking volume, while Online TA has the highest segment cancellation rate.

Management Recommendations

- Strengthen cancellation-risk controls using lead time, customer type and market segment.
- Optimize pricing by hotel type and customer segment using ADR distributions.
- Protect high-risk bookings with targeted deposit, confirmation and reminder policies.
- Prioritize major guest source markets and test localized campaigns for growth markets.
- Compare booking channels on both volume and cancellation exposure before increasing acquisition spend.
- Investigate extreme ADR, lead-time and stay values before removing or capping outliers.
- Maintain an ongoing KPI dashboard covering bookings, cancellations, ADR, lead time, segment mix and source markets.

Analytical Methodology

The workflow standardized text and missing-value tokens, converted reliable numeric fields, checked duplicates and missingness, identified IQR-based outliers, and used targeted exclusion only when a required variable was missing for a particular calculation. Visual analysis used Matplotlib and Seaborn.

Conclusion

The EDA establishes a practical management view of hotel booking behavior. Cancellation risk, booking lead time, ADR, hotel type, customer type and market segment are the most useful dimensions for operational monitoring and commercial decision-making. The findings can serve as a baseline for future forecasting and predictive cancellation modelling.